

Ramez Ashraf Abdelmoniem

Data Engineer | AI & Computer Vision

+201018189406 | rameznasr73@gmail.com | Rehab City, Cairo, Egypt
linkedin.com/in/ramez-ashraf- | github.com/RamezAshraf10-8

PROFFESIONAL SUMMARY

Data Engineer-focused Computer Engineering student with experience in AI, computer vision, cloud computing, and network security. Skilled in Python, SQL, AWS, and machine learning, with strong communication, problem-solving, and self-learning abilities. Seeking a Data Engineer role combining scalable data solutions with hands-on AI and cloud experience.

EDUCATION

The British University in Egypt — Cairo, Egypt

Bachelor of Science in Computer Engineering | 2021–2026

GPA: 3.1/4.0

TECHNICAL SKILLS

Languages: Python, C++, C#, Java, Assembly, SQL

AI, Machine Learning & Computer Vision: YOLOv11 (Ultralytics), PyTorch, OpenCV, Google MediaPipe, Deep Learning, Instance Segmentation & Object Detection, Facial/Eye Landmark Tracking, NumPy, Pandas, Matplotlib

Web & Mobile: React, Flutter, Firebase, Android (Java), TMDB API, EmailJS

Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, RDS, API Gateway, SNS, CloudFront, CloudFormation, DynamoDB, Rekognition, Comprehend), Terraform

Networking & Security: Juniper SRX, Junos OS, IPsec VPN, NAT, Firewall Policies, UTM (Antivirus, Antispam, Web Filtering), High Availability Clustering

Tools & Platforms: MATLAB, ModelSim, Arduino IDE, Blender, Figma, Google Colab, Jupyter Notebook, Pygame

Concepts: Data Structures, Computer Architecture, Microprocessors, Machine Learning, HCI, Assistive Technology, Signal Processing, Penetration Testing, Cloud Architecture

Soft Skills: Problem Solving, Teamwork, Time Management, Communication, Working Under Pressure

INTERNSHIPS AND TRAININGS

Cloud Computing Trainee

ICT Hub | 2025

- Designed and deployed serverless, microservices, and load-balanced architectures on AWS.
- Built projects: AI resume ranker (Comprehend), face recognition alerts (Rekognition + SNS), data pipelines (Athena + QuickSight), and static site hosting (S3 + CloudFront).
- Hands-on experience with core AWS services including EC2, S3, IAM, Lambda, RDS, SNS, API Gateway, CloudFront, and CloudFormation.
- Used CloudFormation and Terraform for IaC; optimized costs via Cost Explorer and Trusted Advisor.

Cybersecurity Intern

ICT Hub | 2025

- Applied penetration testing methodologies (reconnaissance, scanning, exploitation, and reporting).
- Used Kali Linux tools including Nmap and Netdiscover; identified open ports, running services, and vulnerabilities.
- Exploited vulnerabilities using Metasploit: simulated attacks on VMs, established reverse shells, and escalated privileges in a controlled lab.
- Documented findings in structured pen-test reports with remediation steps; completed a capstone simulated pen-test project.

Network Security Intern

BARQ Systems | 2024

- Configured security zones, firewall policies, NAT, and IPsec VPNs on Juniper SRX firewalls using Junos OS CLI and J-Web.
- Implemented UTM features including antispam, antivirus, and web filtering for enterprise environments.
- Set up high availability clustering and monitored real-time network traffic.

AI & Python Trainee

- Studied Python programming and core machine learning concepts.
- Built a chatbot capable of interpreting varied question formats and generating accurate responses from trained data.
- Developed three networked applications: a UDP file transfer tool with server-side renaming, a TCP/SMTP email sender with TLS authentication, and a real-time UDP chat client using multithreading for concurrent communication.

PROJECTS

Smart Vehicle Inspection and Repair Estimator (Graduation Project)

Python, PyTorch, YOLOv11m-seg, OpenCV, Flutter, Firebase

- Developed an end-to-end AI system for automated vehicle damage detection and repair-cost estimation, training a YOLOv11m-seg instance-segmentation model on the CarDD dataset across six damage classes (dent, scratch, crack, glass shatter, broken lamp, flat tire), achieving 0.769 box mAP@0.5 and 0.763 mask mAP@0.5 on the held-out test set.
- Designed a rule-based severity module (minor/moderate/severe) using segmentation mask-to-image area ratios with class-specific thresholds, and a repair-cost estimation module built from survey and workshop pricing data.
- Applied class-specific confidence thresholding and post-processing to filter low-quality detections and improve reliability of user-facing results.
- Built and shipped "FixZone," a cross-platform Flutter mobile app with a Firebase backend, enabling users to scan a vehicle, view damage results, receive severity and cost estimates, download a PDF report, chat with a rule-based assistant, and find recommended repair shops.

Eyelids-Controlled Morse Code Communication System (Bachelor's Dissertation)

Python, OpenCV, Google MediaPipe, NumPy, Pygame, Matplotlib

- Designed and built a real-time assistive-communication system translating intentional eye blinks into text via Morse code, enabling users with severe motor impairments (e.g., ALS, locked-in syndrome) to communicate using only a standard laptop webcam.
- Used Google MediaPipe Face Landmarker to track 12 eye landmarks per frame and compute the Eye Aspect Ratio (EAR), then applied rolling-average smoothing to suppress camera noise and eliminate false-positive blink detections.
- Implemented a timing-based classifier mapping blink duration to dot, dash, and delete gestures, with automatic letter-commitment/word-spacing thresholds and synthesized NumPy/Pygame audio tones for eyes-free feedback.
- Built a delete-gesture error-correction mechanism absent from prior published work, achieving 100% decoding accuracy across five participants versus 62% in the closest benchmark study, while remaining robust to eyeglasses and varying lighting.

Movie Browsing Web App (Etfrag)

React, Firebase, TMDB API, Figma

- Full-stack web app with Gmail auth, multi-profile management, watchlists, language/genre filters, children mode, and EmailJS contact support. Designed high-fidelity UI/UX prototypes in Figma.

Food Ordering Android App

Java, Firebase Realtime Database, GPS

- Native Android app featuring category browsing, voice search, GPS-based location filtering, real-time cart management, and Firebase Authentication.

Retail Management App

C#, SQL

- Desktop application enabling customers to create and edit orders while allowing owners to manage inventory in real time.

Dress Rental Management System

C++ · OOP · Custom Data Structures

- Developed a full CRUD console application managing customers, dress inventory, and rental transactions.
- Designed and implemented template-based linked list and array list from scratch — no STL containers used.
- Features include renting/returning dresses, bill calculation, daily profit tracking, and a 5-dress rental limit per customer.

Network Applications Suite

Python · Socket Programming

- Developed three networked applications: a UDP file transfer tool with server-side renaming, a TCP/SMTP email sender with TLS authentication, and a real-time UDP chat client using multithreading for concurrent communication.